DAY 04 · THEORY HANDOUT

Seaborn: Explore Table Data

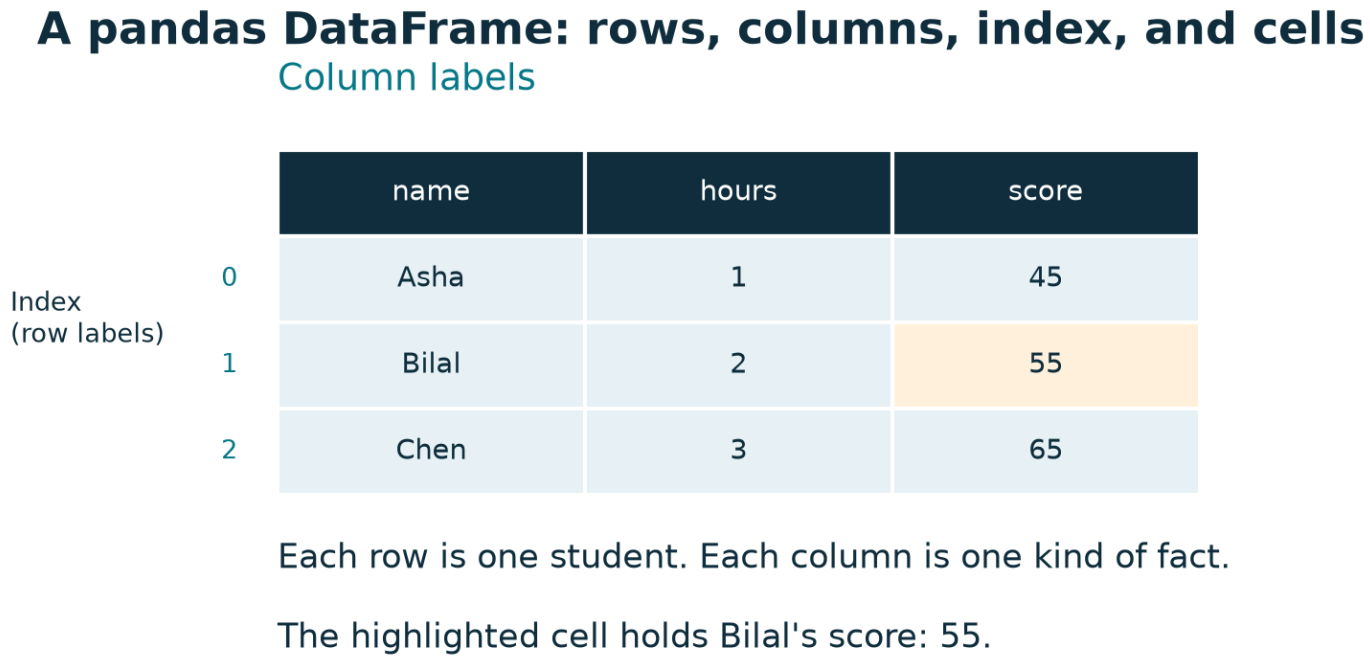
We map named columns to visual features and distinguish raw values from summaries.

KGP Talkie

Scan to subscribe

Laxmi Kant Tiwari · kgptalkie.com

Tidy data and Seaborn



Each student is an observation. Each column is a variable.

Definition

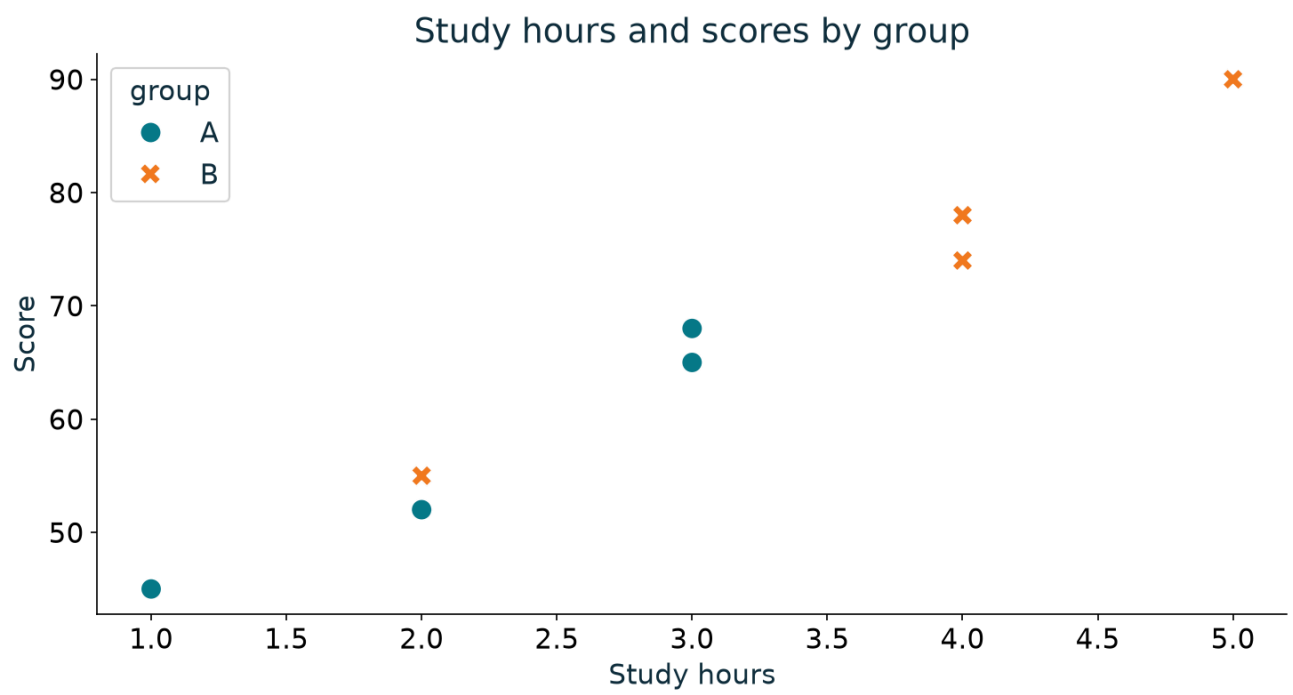
Seaborn is a statistical visualisation library built on Matplotlib. It can read named columns from a pandas DataFrame.

How it works

In **tidy data**, each row is an observation and each column is a variable. Hours and scores are numeric. Group is categorical, meaning it holds a label.

> data=students names the table. x="hours" and y="score" name the columns.

hue and style



The legend links colors and marker shapes to group labels.

Definition

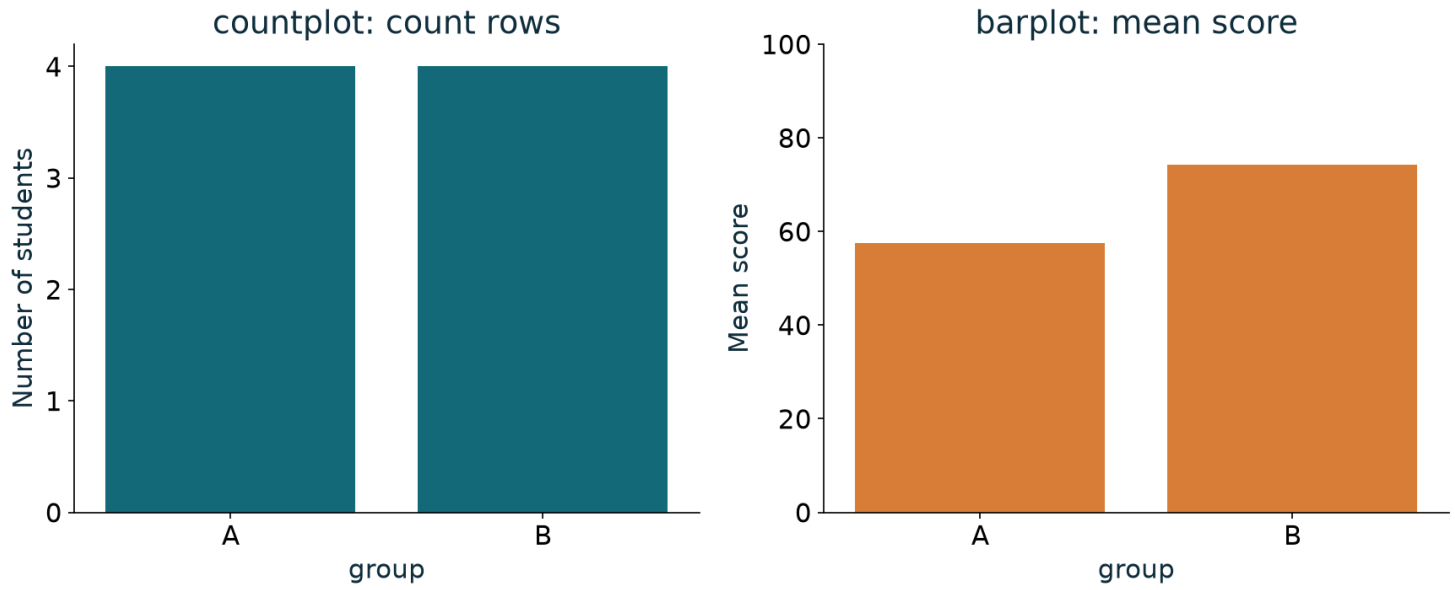
hue maps a variable to color. **style** maps a variable to marker shape. A **palette** is a set of colors.

How it works

We can show two groups on one scatter plot using hue="group" and style="group". Using shape as well as color makes the group difference easier to recognise.

> The point position shows hours and score. Its color and shape show its group.

countplot versus barplot



The left plot counts rows. The right plot calculates mean scores. Their units differ.

Definition

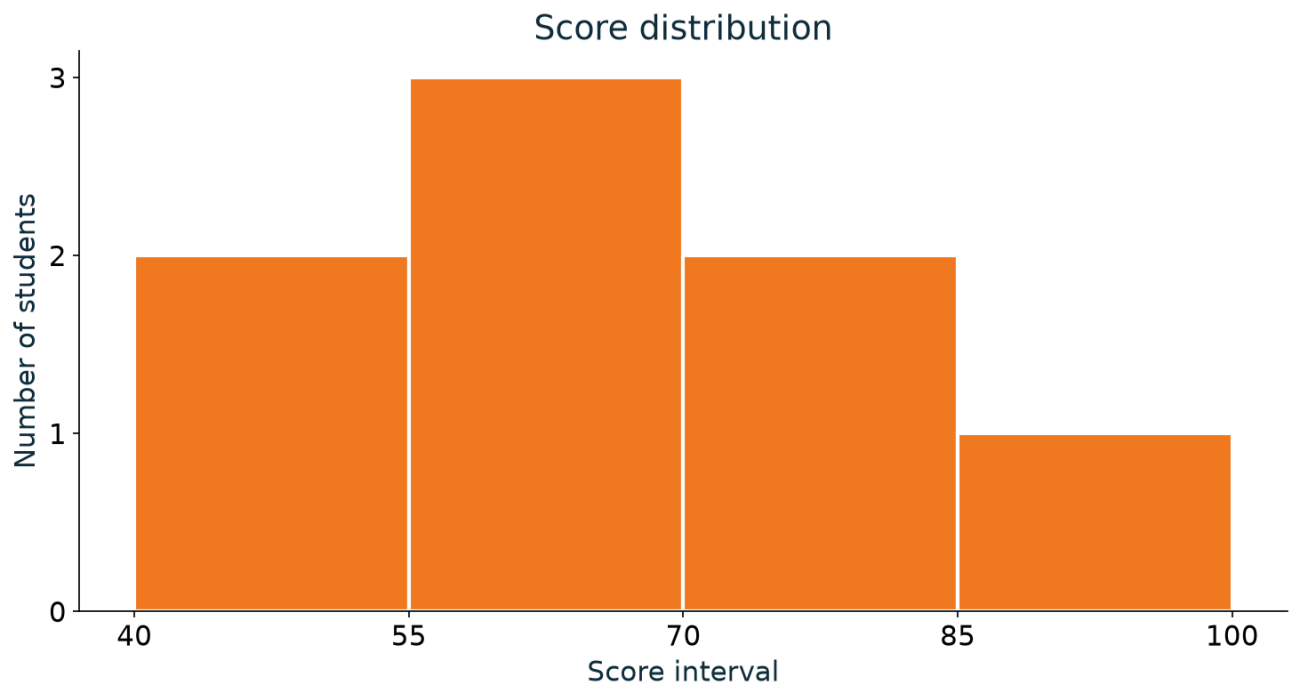
countplot() counts observations for each category. It does not need a numeric y column.

How it works

barplot() summarises y for each category. Its default estimator is the mean. errorbar=None hides uncertainty intervals for this small descriptive example.

> Equal group counts do not imply equal group means. These plots answer different questions.

Histograms and distributions



Each bar counts students in a score interval. The x variable is numeric.

Definition

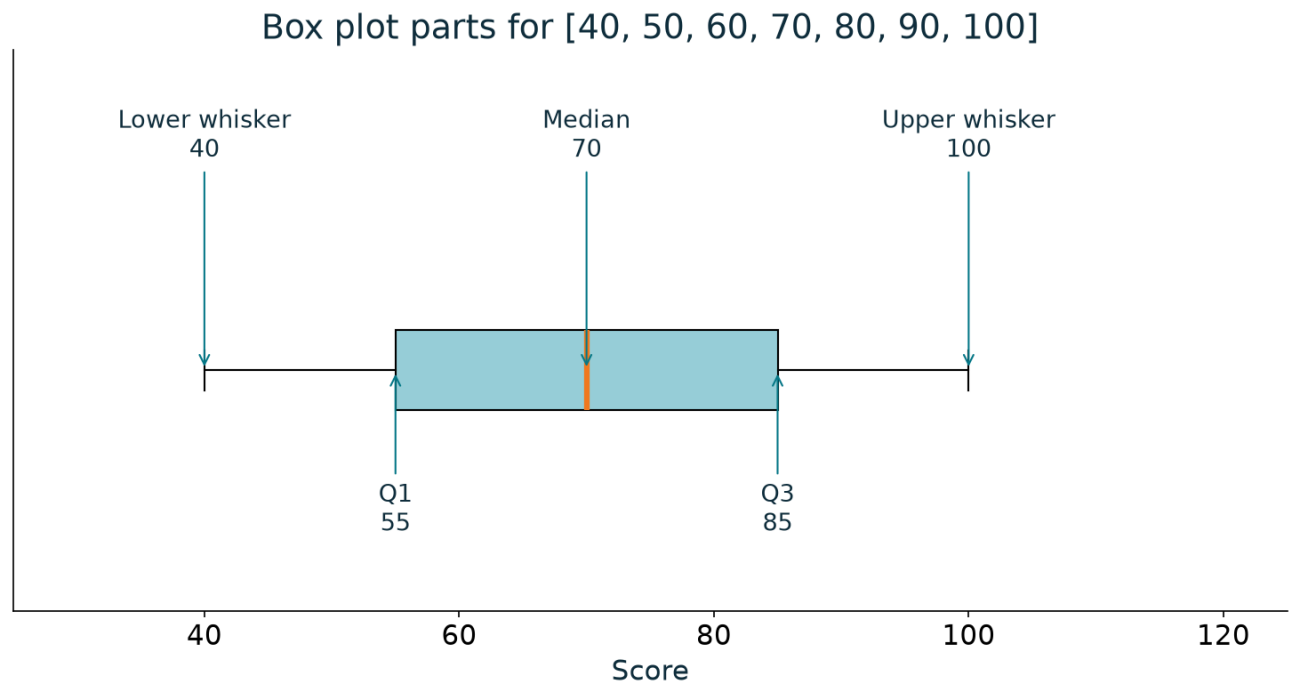
histplot() shows how numeric values are distributed. A distribution describes where values sit and how spread out they are.

How it works

bins can be a count or a list of exact interval edges. Our eight scores use four fixed intervals. Too many bins would leave many empty spaces.

> Try three bins and five bins. The same data can look different when the grouping changes.

Box plot parts



The labeled example uses seven scores. Its median is 70 and its interquartile range is 85 - 55 = 30.

Definition

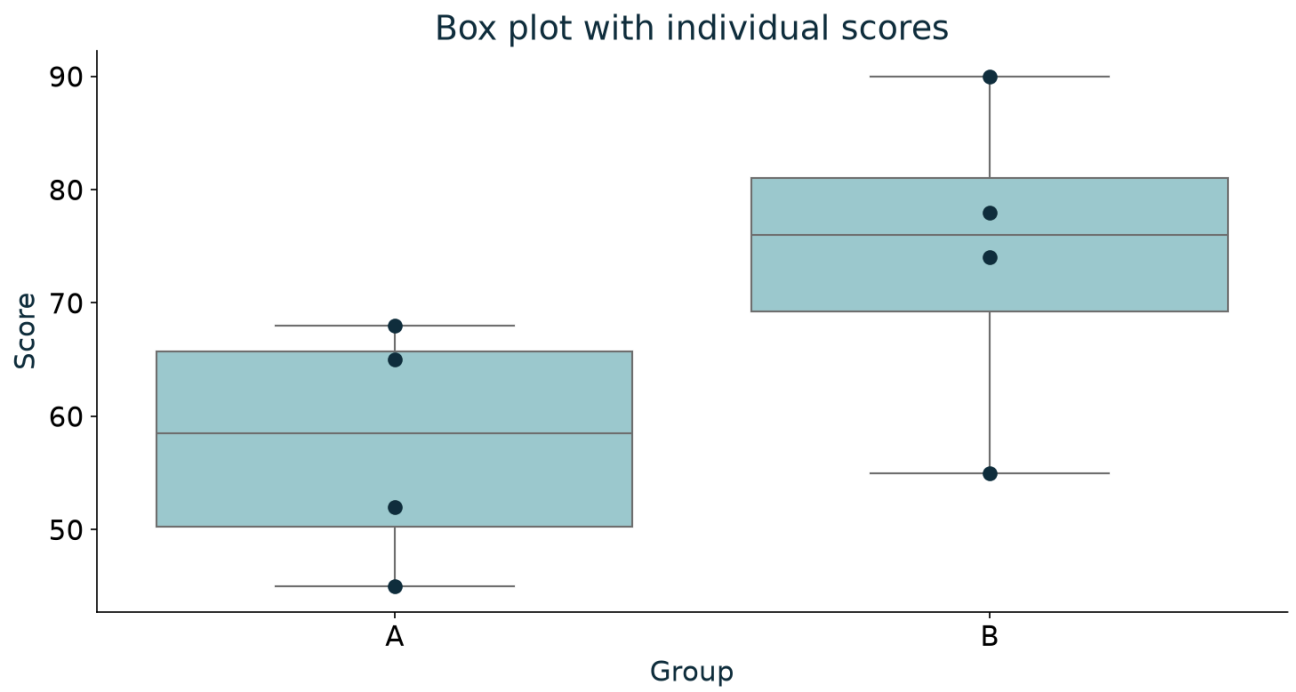
A **box plot** shows a median and spread. The box runs from the first quartile, Q1, to the third quartile, Q3.

How it works

The **IQR** is Q3 minus Q1. Default whiskers reach the furthest values within 1.5 IQR of the box. Points beyond them are possible outliers, not automatically mistakes.

> A box plot summarises data. It does not display every value unless we add the points.

Individual observations



The dark dots display the actual scores alongside their summary boxes.

Definition

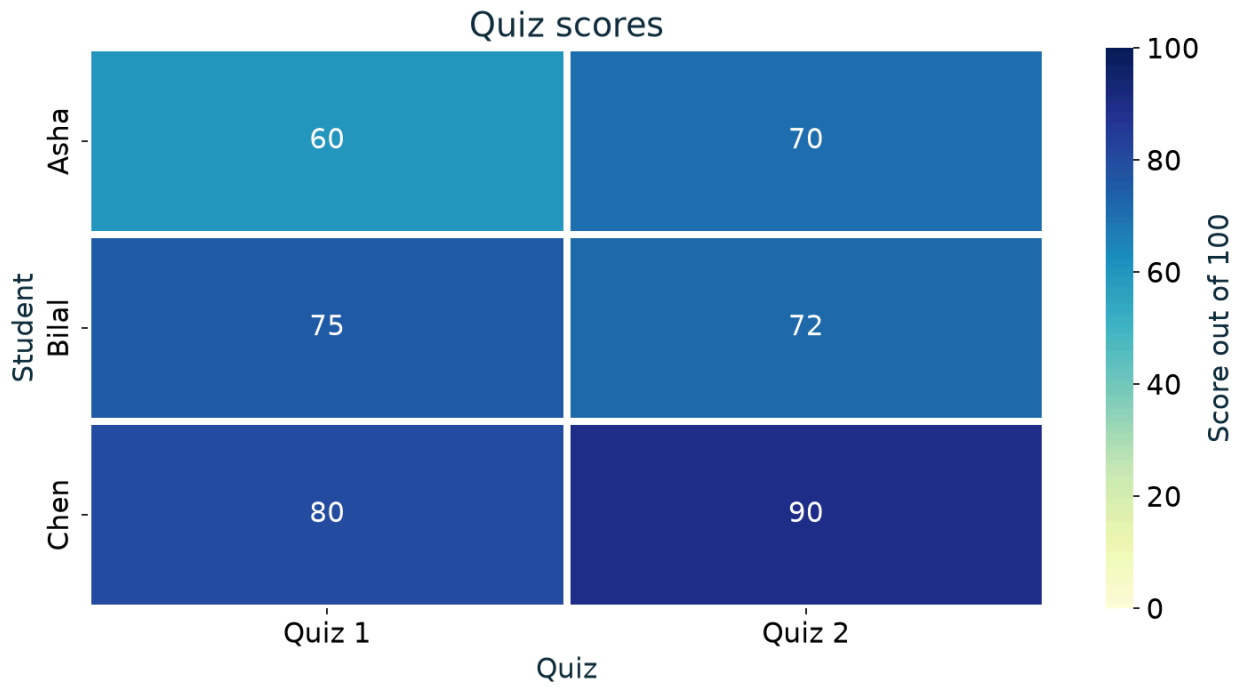
A **strip plot** shows one point for each observation in a category. It can be placed over a box plot.

How it works

Jitter moves points slightly sideways to reduce overlap. In this example we use jitter=False so the points remain at fixed category positions.

There are four students per group. Individual dots make that small sample visible.

Heatmaps and color scales



Each cell is one score. The colorbar gives the shared 0 to 100 scale.

Definition

A **heatmap** maps table values to color. A **colorbar** connects colors to numbers.

How it works

annot=True writes values inside cells. fmt="d" displays integers. vmin=0 and vmax=100 keep a fixed scale. Both quizzes use the same score units.

> Do not compare colors across different units without a suitable shared scaling rule.

Parameters and chart choices

Question	Plot	Useful parameters
Do two numbers vary together?	scatterplot	x, y, hue, style, s
How many rows per category?	countplot	x, color
What is the mean per group?	barplot	x, y, errorbar
How are numbers spread?	histplot / boxplot	bins / x and y
What values are in a table?	heatmap	annot, fmt, cmap, vmin, vmax

An axes-level Seaborn plot can use ax=... to draw into a chosen Matplotlib Axes.

Definition

A **parameter** is a named setting accepted by a function. Our supplied value is its argument. Choosing a parameter changes a specific part of the plot.

How it works

Start with data, x, and y. Add a group mapping when it answers a useful question. Use Matplotlib labels and savefig() to finish and save the figure.

> A theme changes appearance. It does not change the underlying data.

DAY 04 · COMPLETE

Thank You

Follow KGP Talkie for new lessons, course updates and live sessions.

Scan to follow

Instagram
@kgptalkie

WhatsApp Channel
KGP Talkie